

FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[EE4605] Integrated Circuits for RF/Wireless Applications

Assignment 4
Assymmetric DPA

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P_{max}	10W
$P_{back-off\,point}$	-11dB
f_o	1GHz
V_{sup}	10V

Table 1: Data

1 Design steps

1.1 Maximum main and peak current, back-off power and α

At first, I calculate the power at the back off point. In this point only the main amplifier is active so it is equivalent with the power of the main amplifier. Knowing the ratio of P_{max} and $P_{out@backoff}$:

$$P_{out@backoff} = P_{main@backoff} = \frac{P_{max}}{10^{\frac{11}{10}}} \Rightarrow$$

$$P_{out@backoff} = 0.794W$$

I know that the relation between $P_{main@back}$ and $I_{main@max}$ is linear with a factor α . The power follows a quadratic relation so:

$$\alpha = \sqrt{P_{main@backoff}/P_{max}} \Rightarrow$$

$$\alpha = 0.282$$

I can find the maximum main current from the power at back-off:

$$P_{main@backoff} = \frac{1}{2}(a * I_{main@max} * V_{sup}) \Rightarrow I_{main@max} = \frac{2 * P_{main@backoff}}{a * V_{sup}} \Rightarrow$$

$$I_{main@max} = 0.563A$$

Now that I know the $I_{main@max}$ I can find both $P_{main@max}$, $P_{peak@max}$ and $I_{peak@max}$:

$$P_{main@max} = \frac{I_{main@max} * V_{sup}}{2} \Rightarrow$$

$$P_{main@max} = 2.815W$$

$$P_{peak@max} = P_{max} - P_{main@max} \Rightarrow$$

$$P_{peak@max} = 7.184W$$

$$I_{peak@max} = \frac{2 * P_{peak@max}}{10} \Rightarrow$$

$$I_{peak@max} = 1.437A$$

1.2 Transconductance

The device schematics suggest that the current of each amplifier is following the expression:

$$I = gm * Vin \Rightarrow$$

$$gm = \frac{I}{Vin} \quad (1.1)$$

In order for the devices to support the maximum main and peak current the transconductance of each of them should be:

$$gm_{main} = \frac{0.567}{1} \Rightarrow$$

$$\boxed{gm_{main} = 0.563S}$$

$$gm_{peak} = \frac{1.436}{1} \Rightarrow$$

$$\boxed{gm_{peak} = 1.437S}$$

1.3 R_o and R_L

At the back-off point we got:

$$I_{peak} = 0 : \quad Z'_1 = R_L \quad \text{and} \quad Z_1 = \frac{R_o^2}{R_L}$$

The impedance that the main amplifier sees is:

$$Z_{main@back} = \frac{V_{sup}^2}{2 * P_{main@back}} = \frac{100}{2 * 0.7943} \Rightarrow$$

$$\boxed{Z_{main@back} = 62.949\Omega}$$

By substituting this value to Z_1 we got:

$$62.9485\Omega = \frac{R_o^2}{62.9485} \Rightarrow$$

$$R_o = \sqrt{62.9485 * R_L} \quad (1.2)$$

In the case of the maximum output power the voltage swing has an amplitude of V_{sup} , so:

$$R_L = \frac{V_{sup}^2}{2P_{max}} = \frac{100}{2 * 10} \Rightarrow$$

$$R_L = 5\Omega$$

Substituting the R_L in equation 1.2 we can find R_o :

$$R_o = 17.74\Omega$$

1.4 Output impedance matching

As it is usual the load of the amplifier is 50 Ohm. However, after the optimization that was performed I found that the Doherty amplifier needs to see a 5 Ohm load at its output.

For this reason I place a $\lambda/4$ impedance transformer just before the load. The impedance of this transformer needs to be:

$$Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta l)}{Z_o + jZ_L \tan(\beta l)} \Rightarrow 5 = Z_o \frac{jZ_o \tan(\beta l)}{j50 \tan(\beta l)} = \frac{Z_o^2}{50} \Rightarrow Z_o = \sqrt{50 * 5} \Rightarrow$$

$$Z_o = 15.81\Omega$$

Since we talk about transmission lines, I have to stress that there are another two transmission lines at the biasing of the amplifier. The biasing nodes of the circuit are considered short, and need to provide an open for the fundamental. For this reason, I apply an electrical length of 90 degrees with an impedance of 50 Ohm (in this case the impedance is irrelevant).

2 Deliverables

2.1 Amplifier Schematic

All the design steps were implemented in the given template as shown below:

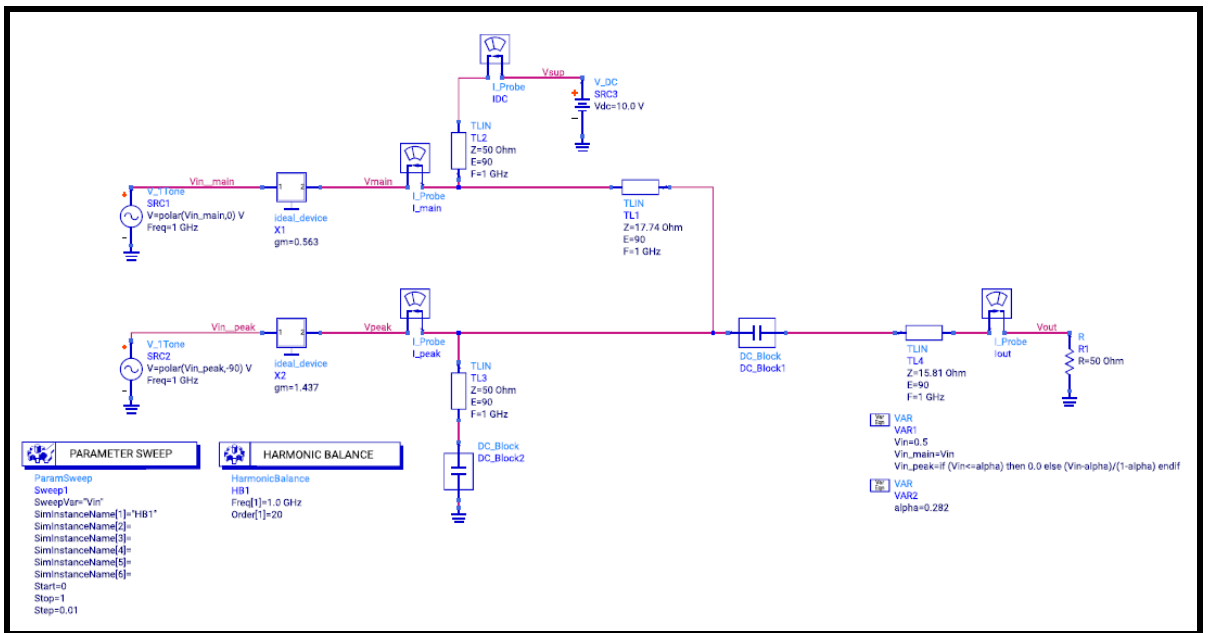


Figure 2.1: Amplifier Schematic.

For all the graphs except the time domain waveforms only the fundamental has been plotted. In the case of V_{main} and V_{peak} plotting the output waveform with the harmonics as well, do not show any visible difference. This is because the even harmonics are shorted from the $\lambda/4$ lines at the biasing points and the impedance seen by the odd harmonics are really small because the matching network is designed for the fundamental.

2.2 Main and peak current versus input voltage

In figure (2.2), the main amplifier is active at the time an input voltage is applied, while the peak amplifier start to operate at 0.282V. The activation of the peak amplifier means that the magnitude of the output voltage of the main amplifier is 10V and therefore the current is now drawn from the peak amplifier as well. Since more current can now be delivered to the load, the applied voltage is getting bigger, which translates for the main amplifier as a decrease of the impedance. This is because just after the main amplifier, there is a $\lambda/4$ line that inverts the effect of the bigger current. We would expect actually to see a bigger impedance if it wasn't for the $\lambda/4$ line. This way, we prevent any clipping of the main amplifier.

The parameter α was calculated from the specifications and indicates at which back-off point the peak amplifier needs to turn on. Since the $V_{in,max}$ is 1V, at 0.282V the peak amplifier needs to be activated.

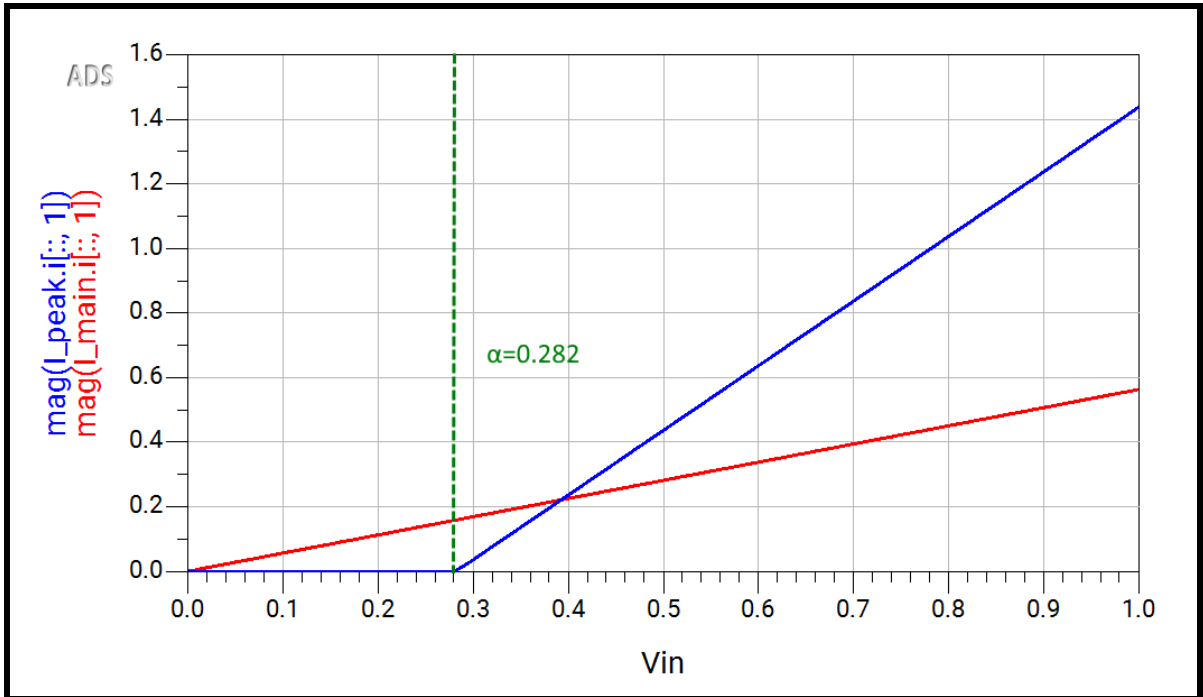


Figure 2.2: I_{main} and I_{peak} over V_{in} .

2.3 Main and peak voltage versus input voltage

In figure (2.3), the V_{main} reaches 10V amplitude at $V_{in}=0.282V$, while the V_{peak} reaches this value at the maximum of V_{in} as it was expected.

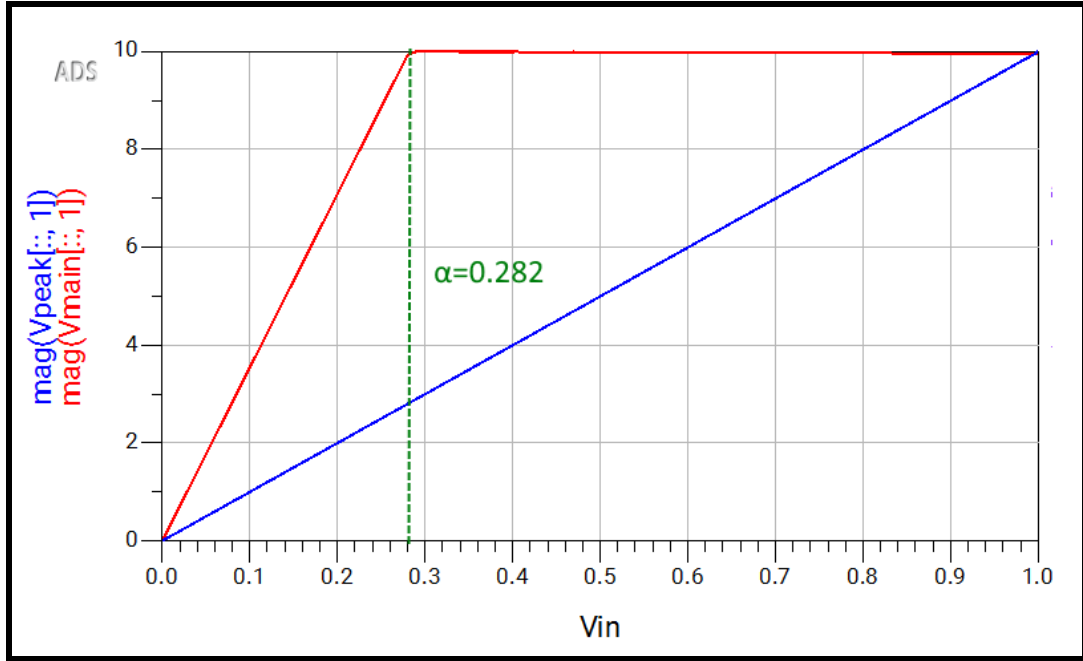


Figure 2.3: V_{main} and V_{peak} over V_{in} .

2.4 Loading conditions over input voltage

In figure (2.4), the impedance seen by the main amplifier before 0.282V is $Z_{main@backoff} = 62.949\Omega$ as I calculated during the design process. After the back-off point, the impedance seen by the main amplifier is decreasing, validating the previous analysis. Moreover, the impedance of the peak amplifier starts from a really high value because when the peak amplifier turns on it draws a small current, while the voltage swing to the load is already high due to the main amplifier.

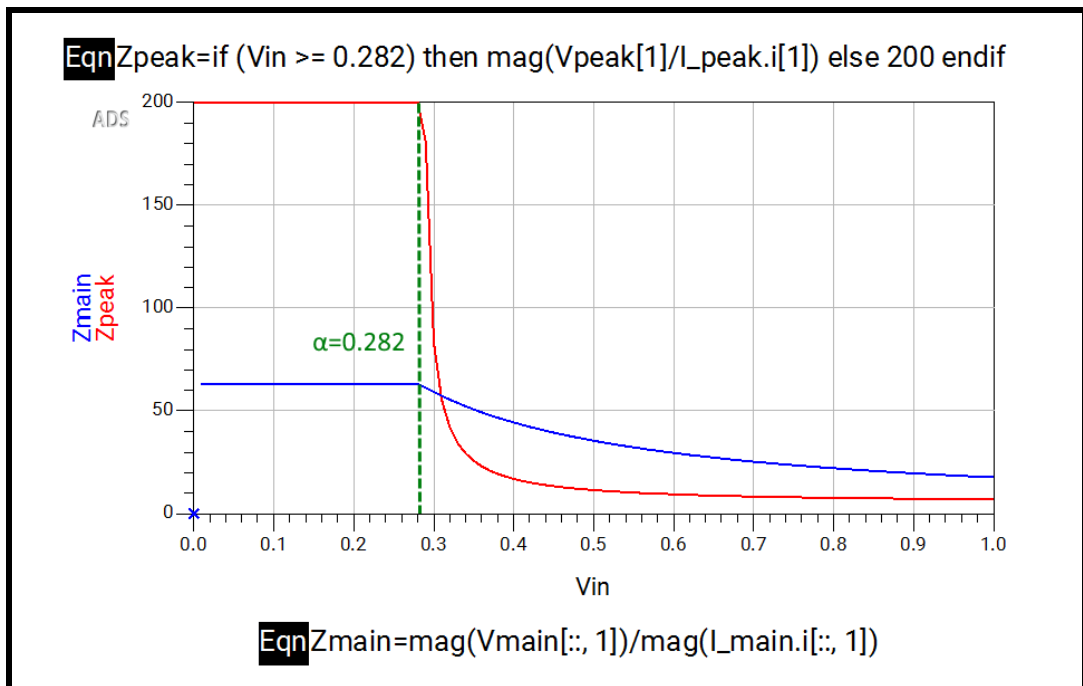


Figure 2.4: Z_{main} and Z_{peak} over V_{in} .

2.5 Efficiency versus output power

According to the lecture notes, the efficiency is described:

$$\eta = \frac{\text{RF output power}}{\text{consumed DC power}} = \frac{P_{out}}{P_{DC}} \quad (2.1)$$

The DC power is:

$$P_{DC} = V_{sup} * I_{DC} \quad (2.2)$$

The output RF power is the power dissipated in the output load:

$$P_{out} = \frac{1}{2} \Re\{V_{out} * I_{out}^*\} \quad (2.3)$$

for which the dBm representation is:

$$P_{outdBm} = 10 \log\left(\frac{P_{out}}{1mW}\right) \Rightarrow$$

$$P_{out} = 10 \log\left(\frac{1}{2} \Re\{V_{out} * I_{out}^*\}\right) + 30 \quad (2.4)$$

In figure (2.5), the output RF power has been normalized according to the maximum value to show that the back-off point is located at -11dB from the maximum power, which is in agreement with the specifications.

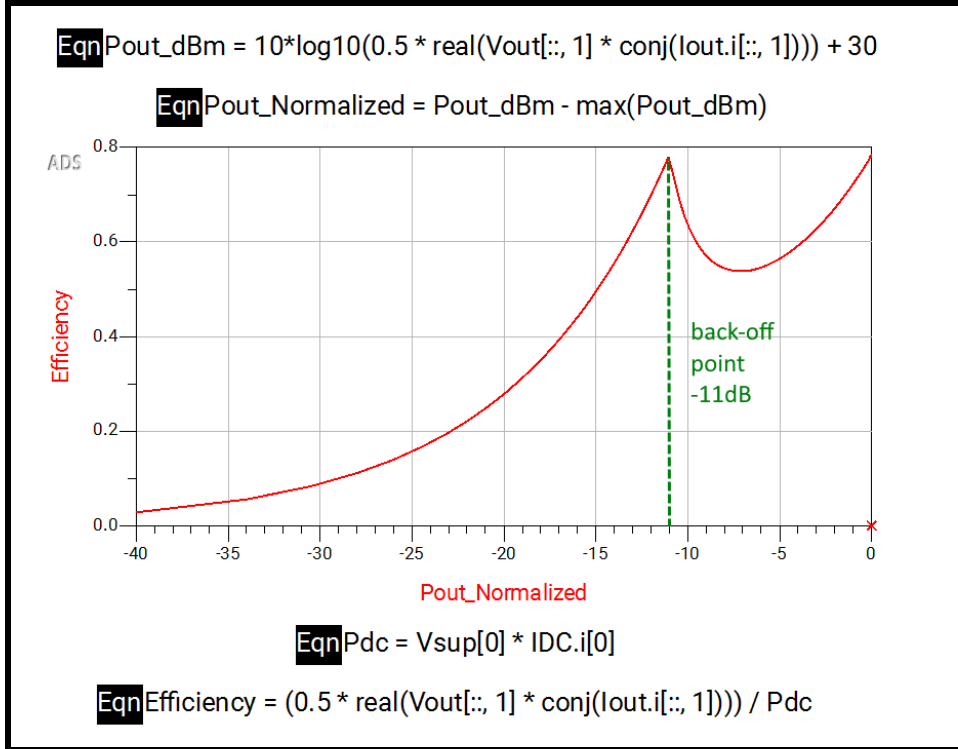


Figure 2.5: Efficiency versus output power.

2.6 Time domain waveforms V_{main} and I_{main}

In the following figures, V_{main} and I_{main} are plotted. I have plotted both of them in three phases; before the 0.28V where the voltage swing at the output of the main amplifier is still scaling, at the back-off and at maximum input voltage. As it is expected the voltage swing at the back-off and at maximum input voltage is the same, due to the effect of the peak amplifier as described before.

We observe a perfect sine wave for the V_{main} , but this is because the input voltage range is just high enough to not clip both the peak and the main amplifier. As we have seen before, for $V_{in}=1V$ the magnitude of the peak output voltage has also reached 10V, so if we increase a bit more the voltage, then the amplifiers will clip. Also even though we plot all the harmonics, there is no visible distortion for the reason I described in paragraph 2.1.

Moreover, regarding the reason that the voltage swing goes higher than the power rails. When the input voltage is enough to drive the device, then the device conducts and amplifies the signal. When it is below the threshold voltage of the device, it stops conducting, but the RF choke at the biasing node fights the current change by increasing the voltage at the output of the amplifier. In conclusion, it is the RF choke that provides the extra voltage when the device is not conducting.

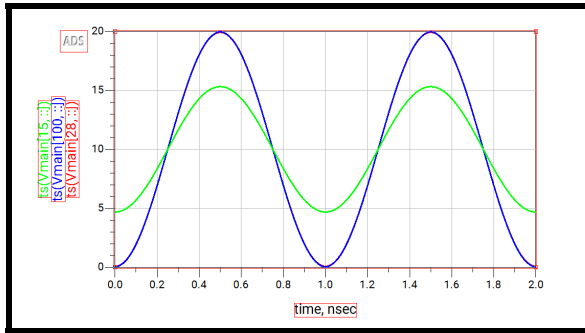


Figure 2.6: V_{main} over time.

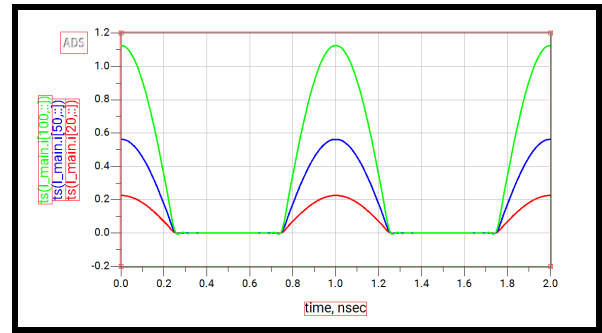


Figure 2.7: I_{main} over time.

In figure (2.8), I plot the voltage and current at the same plot in the case of $V_{in}=1V$ to show the power that the device dissipates.

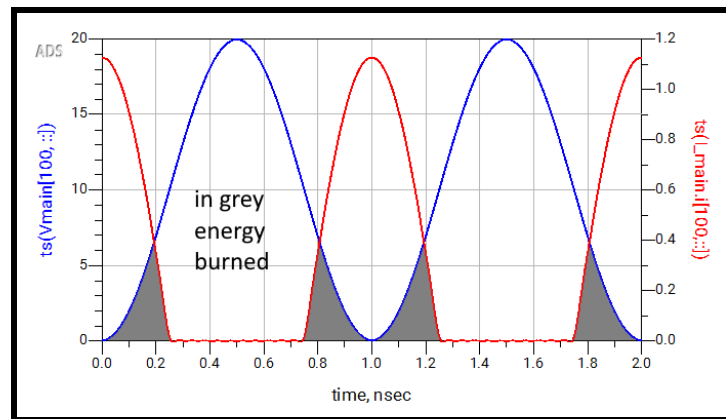


Figure 2.8: Combined plots.

2.7 Time domain waveforms V_{peak} and I_{peak}

In the case of the peak amplifier, the magnitude of the output voltage is increasing linearly with V_{in} , as it is presented in the following figures. The amplitude only peaks for $V_{in}=1V$. The logic behind the waveforms remain the same with the main amplifier, but we can clearly see a phase difference of 90 degrees. This was implemented by the test bench so the two output signals of the amplifier would add up at the load.

Regarding the current plot, the conduction angle remained the same, meaning that technically it is a Class-B amplifier (both of the amplifiers have a conduction angle of 180 degrees). A class C requires more voltage to be activated than the class B, thus it might be a better choice for a peak amplifier; however, in our case the switching of the amplifiers is ideal. A class-B though still burns more power, since it is not that efficient as a class C. Finally, I don't include a current plot for $V_{in}<0.282$ since the peak amplifier is not activated.

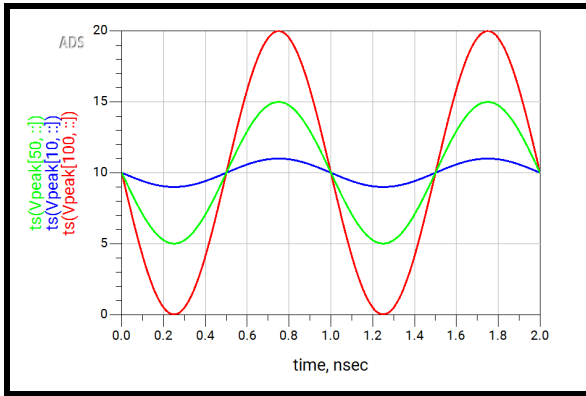


Figure 2.9: V_{peak} over time.

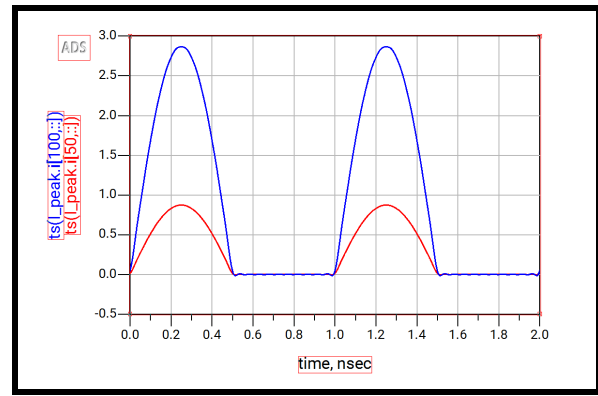


Figure 2.10: I_{peak} over time.

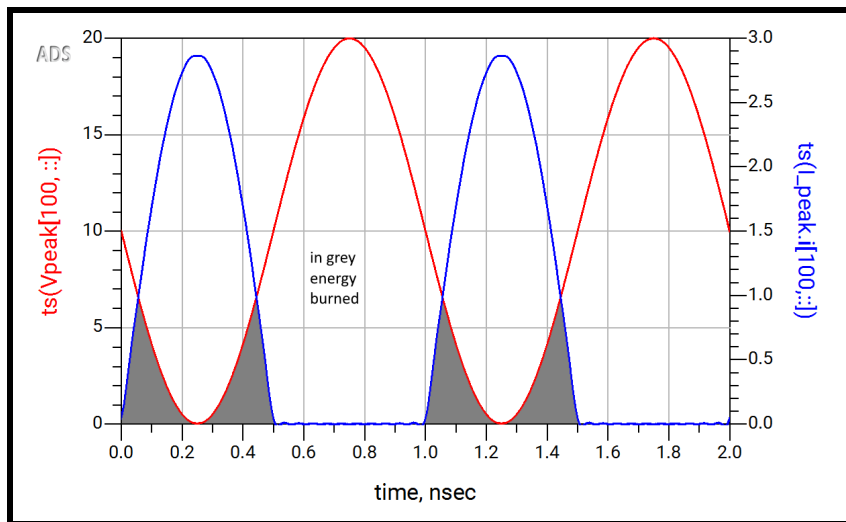


Figure 2.11: Combined plots.

2.8 Loadlines

In figure (2.12), the loadlines for four different power levels are presented; maximum power, back-off point, smaller than back-off and greater than back-off.

At first, for maximum power we can observe that the loadline is the steepest. That means that the conductance is the largest out of all cases and the impedance seen by the main amplifier is the smallest. This complies with the fact that at maximum power the peak amplifier is providing the most current and that translates to the smallest impedance that the main amplifier sees. Furthermore, we can see that the amplitude of the output signal is 10V, just like the two other loadlines for the back-off and the intermediate point. Because, as I discussed after the back-off point the main amplifier has a steady voltage swing of 10V amplitude and the extra current is provided by the peak amplifier.

Secondly, we observe that the back-off point has the shallower slope. That means that the conductance is the smallest, so the impedance is the biggest. This is logical as well, because the peak amplifier has not activated yet and the main amplifier sees the impedance without any decrease due to the peak amplifier. The intermediate point has a slope between those two because the peak amplifier is activated but still provides less current than the maximum power case.

Lastly, for $V_{in} < 0.282V$ the peak amplifier is not activated. That means that the slope is the same with the back-off point and the impedance seen by the main amplifier doesn't change. However, the voltage swing is smaller since we have not reached the peak efficiency of the main amplifier. The fact that all the loadlines have zero current at 10V is because the device has switched off. The voltage after that is increasing due to the RF choke as described previously.

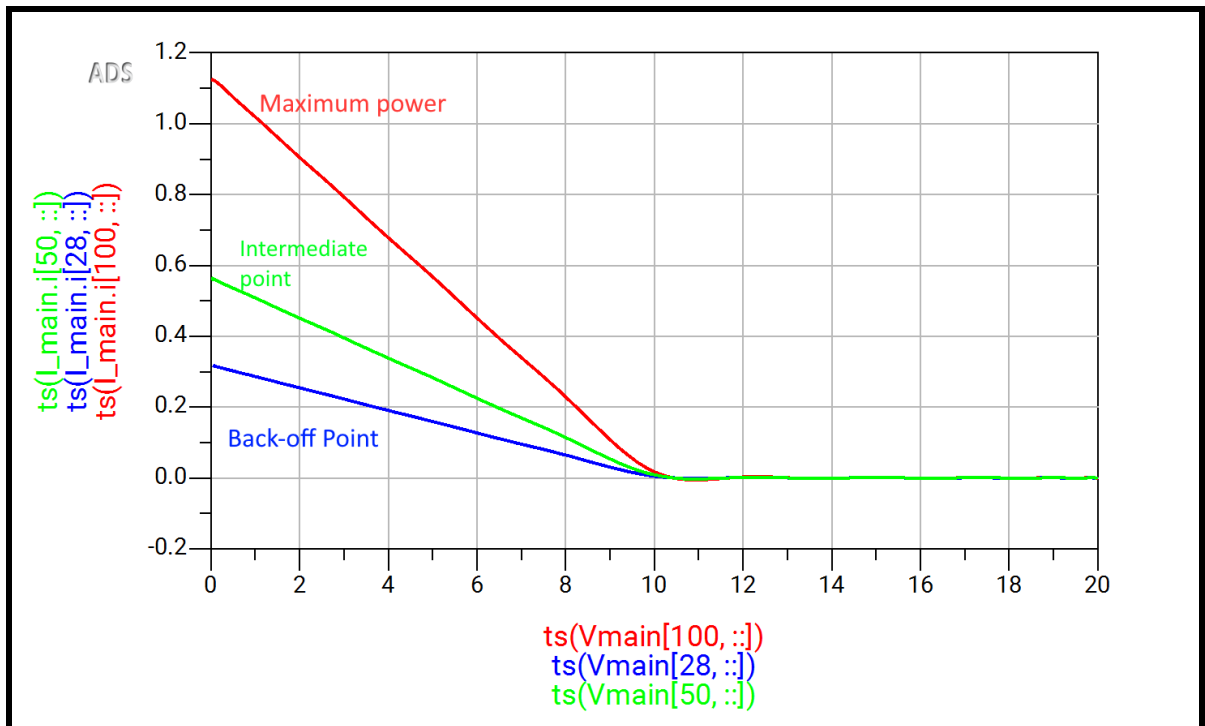


Figure 2.12: Loadlines from back-off to maximum power.

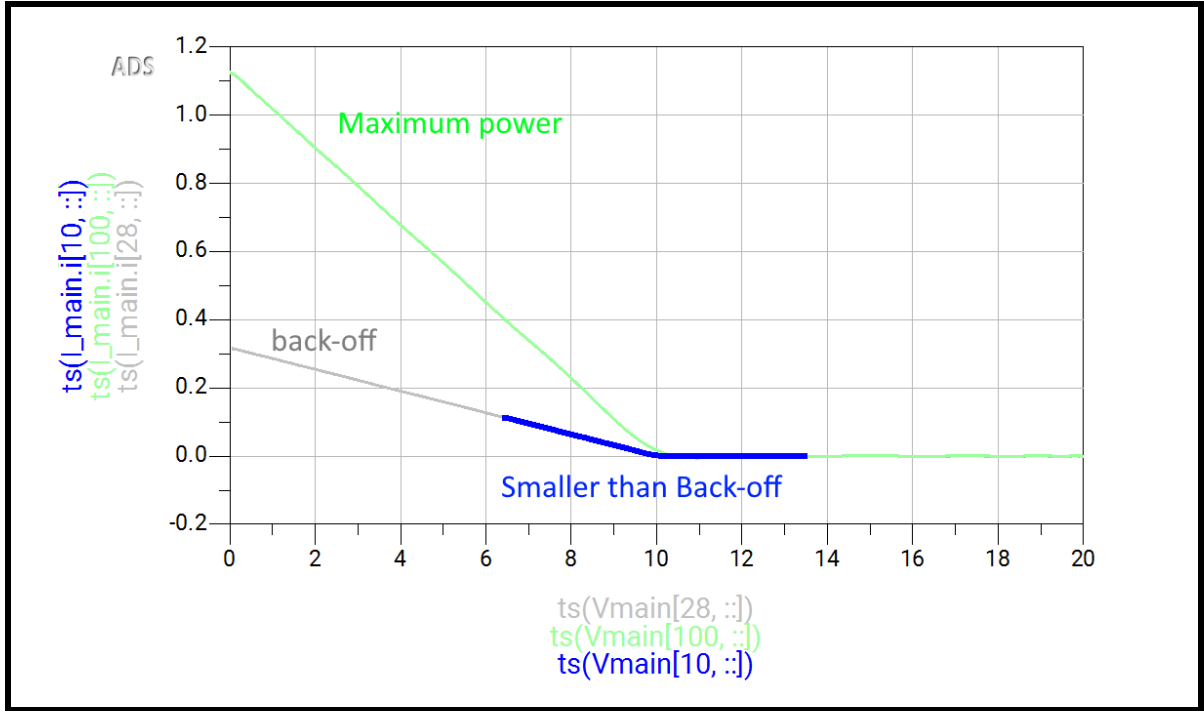


Figure 2.13: Loadline for power smaller than back-off.

2.9 Efficiency suggestions

Firstly, I assume that the given specification are especially chosen to bring the maximum efficiency at the same point where the average power is positioned. With this assumption, any movement of the efficiency point by making the Doherty more symmetric or asymmetric will not have a positive effect to the overall efficiency.

A suggestion could be to use a three-way Doherty. The idea is that we add an extra peaking amplifier, which switch in a different point than the other. From the figure (2.5), we can observe that after the first peak of the efficiency there is a valley. The three-way Doherty, will add an extra peak in this region, so the overall efficiency in high powers will increase.

Another suggestion would be to make the main amplifier class-F. Right now, the odd harmonics at the output of the main amplifier see a really small impedance. We could change our output matching for the odd harmonics, in order for them to see a really large impedance; for example, we could use a resonator for the fundamental frequency at the output. This way, the odd harmonics will shape the signal at the output of the main amplifier into a square wave. This will result in smaller overlapping of the current and voltage (figure 2.8) and therefore the power burned on the device will decrease (higher efficiency).